

Network Troubleshooting Guide

Structured diagnosis for connectivity, WiFi, and performance

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The Layered Method (Work Up the Stack)

- 1 Physical: link lights, cable integrity/reseat, correct port, PoE budget, switch port errors (CRC = bad cable/duplex).
- 2 Addressing: valid IP? (ipconfig) — 169.254.x.x = DHCP failure → check scope exhaustion, VLAN tag, DHCP service. Duplicate IP? Wrong gateway?
- 3 Local network: ping gateway → other LAN hosts. Gateway fails = local infra issue; others fail = switching/VLAN/isolation.
- 4 DNS vs. connectivity: ping 8.8.8.8 vs. nslookup google.com — IP works but names don't = DNS; fix resolver, don't chase 'internet down'.
- 5 Path/WAN: traceroute; modem/ONT status; ISP outage line; failover status if dual-WAN.

Common Scenarios

Scenario	Likely Causes & Actions
Whole office down	Power event → verify UPS held core gear; ISP outage → modem lights + ISP status, open ticket with LOA authority (BVT-OPS-030); firewall hang → controlled reboot, check logs after
One VLAN/segment down	Switch uplink, VLAN config change, DHCP scope for that VLAN, STP event — check switch logs and recent changes first
Intermittent drops	Duplex mismatch, failing cable/SFP, loops (broadcast storms), DHCP conflicts, WiFi roaming — get evidence: interface error counters and continuous ping with timestamps
Slow internet, LAN fine	Speed test at the firewall vs. behind it; saturation (identify top talker in firewall reports — backup or cloud sync mid-day?); ISP degradation; QoS misconfig
Can't reach one site/app only	DNS record, upstream provider outage (check status page), geo/security blocking, MTU issues on VPN paths (test with reduced MTU)
VPN won't connect	Confirm which phase fails: auth (creds/MFA), IKE (config mismatch/blocked ports), or route/DNS after connect — logs on the concentrator tell you

WiFi Specifics

- Signal ≠ quality: check retry rates, channel utilization, and co-channel interference — not just bars.
- One area weak: AP placement/power; verify roaming thresholds; add AP only after a survey, not by default.
- Random disconnects: DFS channel radar hits (move off DFS if repeated), band-steering misbehavior, driver issues on a specific device model.
- Guest network must remain isolated from LAN (verify after any firewall change) — a policy requirement, not just a preference (BVT-POL-044 PR.IR).

Document as You Go

Every network incident updates BVT-OPS-026 documentation: what failed, what fixed it, what changed. Root causes marked 'unknown' get a scheduled follow-up, not a closed ticket — intermittent issues always return.